

either be vacuum or pressed-down. A vacuum fixture gives better signal reading versus the pressed-down type; although, the former is more expensive due to its high manufacturing complexity. Some other benefits of ICT are:

- ICT takes less time to perform the test. For example, if a flying probe takes twenty minutes or so, ICT for the same test might take a minute or so.
- It checks and detects shorts, opens, solder bridges, missing components, wrong-value components, wrong polarities, all manufacturing defects, design faults and flaws, component failures and current leakages in the circuitry.
- Testing platform is available for most testing needs on Windows as well as UNIX.
- Operating environment and test development interface are based on standards for open system offerings. ICT also enables fast integration into an OEM customer's existing processes.
- However, a costly fixture is required as ICT covers hundred per cent testing so that all power and ground shorts are detected.

Role of software in ICT

An in-circuit tester can be programmed easily by taking files from the PCB layout to generate much of the required program. To speed up the process of testing, there are different software available that can run required tests on the assembly. Software can be written for each board type to be tested.

The software instructs the test system which tests to perform, at what points, and details of correct and defective component criteria. It runs tests designed to identify any potential problems by measuring electrical parameters, powering on and off certain areas, and flagging any results that indicate an issue.

Automation of the testing process enables manufacturers to check each finished product rapidly and consistently. This also increases speed and overall quality levels. Some parameters that the regular ICT setup checks include:

Error detection. A more tightly-populated PCB is more exposed to damage during the assembly process. An ICT detects and isolates any open or short-circuits on the PCB board.

Part placement. An ICT routinely checks for improperly placed or substituted components.

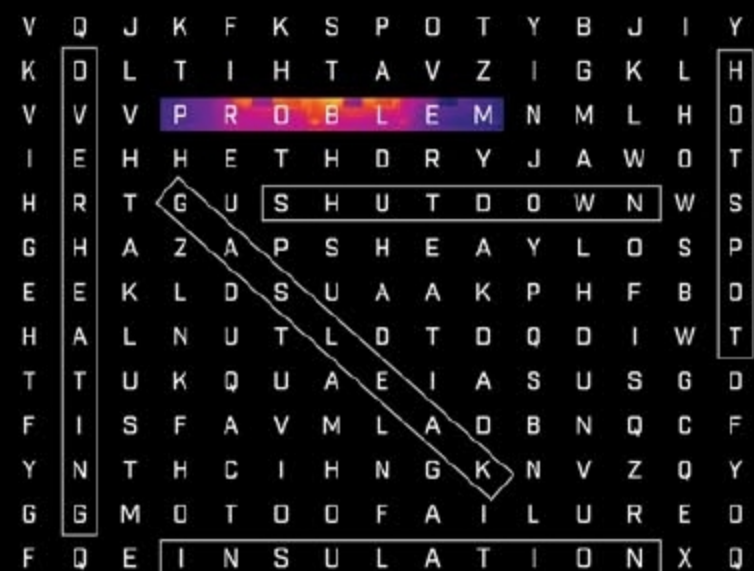
Electrical requirements. It runs the power signal to check voltage levels, resistance and capacitance measurements at different nodes of the board to determine if they meet design specifications.

Component condition. By measuring component reaction at specified voltages and currents, designers can quickly locate and replace damaged components.

Once testing is completed, a report is generated that includes points about tested, partially-tested and untested components, details of fixture wiring and probe types used, and software employed, so that the PCB fabricator can closely analyse the report and make appropriate changes accordingly. **EFY**



PROBLEMS ARE ALWAYS HIDDEN, AND WE HELP YOU FIND THEM



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